

# Unlinked Gamma regression: exact identifiability and optimal cumulant order

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## Abstract

Unlinked linear regression asks which coefficient vectors can produce a prescribed response distribution from a known predictor distribution. For a fixed dictionary of independent, centered, standardized Gamma predictors with arbitrary positive shapes, we characterize equality of response laws by a finite measure recording the total shape at each nonzero signed effective scale. This gives all coefficient fibers as finite allocation problems and shows that minimality relative to the given dictionary is equivalent to full support. A three-predictor example therefore contradicts the signed-permutation conclusion of a conjecture of Balabdaoui, Slawski and Steffani while retaining finite identifiability. We give a necessary and sufficient arithmetic condition for every minimal representation to be identifiable up to signs and permutations. The condition requires equal subset sums of shapes to have the same constituent shape multiset; only equal-shape permutations then occur. Cumulants through order  $2d + 1$  determine the response law and its coefficient fiber, and this consecutive-order cutoff is optimal uniformly over dictionaries with signed coefficients. We also give exact reconstruction, a one-Gaussian extension, and an extension to specified analytic convolution families, including Poisson predictors.

**Keywords:** unlinked regression; identifiability; Gamma convolution; subset sums; cumulants; Prony reconstruction.

## 1 Introduction and model

Suppose that the distribution of a predictor vector  $X$  and the distribution of a response  $Y$  are known, but the links between their observations are unavailable. In the noiseless linear model, the population problem is to determine

$$\mathcal{B}_X(Y) = \{\beta \in \mathbb{R}^d : Y \stackrel{d}{=} \beta^\top X\}.$$

This set can be a singleton, finite with more than one element, or infinite. These possibilities must be distinguished from uniqueness up to a prescribed group of symmetries.

Balabdaoui, Slawski and Steffani [3] discuss these distinctions and, in the concluding conjecture of their version 1, propose signed-permutation identifiability for independent unit-variance predictors with at most one Gaussian component, assuming that the representation is minimal. Here minimality excludes a representation using a proper subvector of the given predictors. Their paper also identifies convolution as an obstruction and proves a positive mixed Gamma–Gaussian result under distinct labeled subset sums of the Gamma shapes.

We analyze the fixed Gamma dictionary completely. Throughout Sections 1–5, let  $d \geq 1$ , let  $\alpha_1, \dots, \alpha_d > 0$  be known, and let

$$G_i \sim \text{Gamma}(\alpha_i, 1) \quad \text{independently,} \quad X_i = \frac{G_i - \alpha_i}{\sqrt{\alpha_i}}. \quad (1)$$

The Gamma parameters are shape and rate. Each  $X_i$  has mean zero and variance one. Set

$$Y_\beta = \beta^\top X, \quad a_i = \frac{\beta_i}{\sqrt{\alpha_i}}, \quad \mathcal{B}(\beta) = \{\gamma \in \mathbb{R}^d : Y_\gamma \stackrel{d}{=} Y_\beta\}. \quad (2)$$

Coefficients can be positive, negative, or zero. The dictionary, including its labels and shapes, remains fixed.

### 1.1 Contribution and relation to classical results

The analytic invariant used below is closely related to the classical Thorin measure of a positive Gamma convolution [4]. For positive  $a_i$ , the uncentered sum  $\sum_i a_i G_i$  has Thorin measure  $\sum_i \alpha_i \delta_{1/a_i}$ . We use the signed scales  $a_i$  directly and give a short proof based on rational logarithmic derivatives that also handles centering and negative coefficients. The distinction between this aggregate invariant and its allocation to a fixed list of predictors is central to the identifiability results.

The statistical context includes deconvolution-based estimation with unmatched data. Azadkia and Balabdaoui [1] study coefficient estimation under identification and a distance-to-solution-set formulation in nonidentifiable cases. Balabdaoui, Di Noia and Durot [2] study recovery of the latent linear-predictor distribution in unlinked linear models, obtaining parametric-rate results under their assumptions. Our population classification makes the coefficient ambiguity explicit for a known Gamma dictionary. Recovery of a latent distribution and identification of its coefficient vector are different targets. Exact cumulant sufficiency alone does not validate a sample-cumulant estimator or transfer these statistical rates to every Gamma configuration.

Finite-atomic moment reconstruction is classical Prony theory [5]. Our reconstruction argument applies that theory after weighting each scale by its square. The additional lower-bound construction respects a single fixed Gamma dictionary on both sides. Neither the rational-function method nor the abstract moment-reconstruction method is presented as new. The literature comparison accompanying this manuscript records bounded uncertainty about historical priority of the particular allocation criterion and optimal-cutoff formulation.

Theorem 2.1 identifies the invariant; Theorem 2.2 gives every fiber and characterizes minimality. Section 3 verifies the minimal counterexample. Theorem 4.1 gives the sharp arithmetic replacement for the conjectured conclusion. Theorems 5.1 and 5.2 establish the optimal consecutive cumulant cutoff. Section 6 states the precise scope of the extensions.

## 2 The aggregate measure and all coefficient fibers

Define the finite positive measure on  $\mathbb{R} \setminus \{0\}$

$$\mu_\beta = \sum_{i: a_i \neq 0} \alpha_i \delta_{a_i}. \quad (3)$$

Repeated scales are combined by adding their shape weights. The measure is positive even when its support contains negative numbers.

**Theorem 2.1** (Exact distributional invariant). *For all  $\beta, \gamma \in \mathbb{R}^d$ ,*

$$Y_\beta \stackrel{d}{=} Y_\gamma \iff \mu_\beta = \mu_\gamma.$$

*Proof.* There is an open interval about zero on which all factors  $1 - a_i t$  are positive and the moment-generating function exists. On that interval the cumulant-generating function and

its derivative are

$$K_\beta(t) = \sum_i \alpha_i \{-a_i t - \log(1 - a_i t)\}, \quad (4)$$

$$K'_\beta(t) = \sum_{i:a_i \neq 0} \alpha_i \left\{ \frac{a_i}{1 - a_i t} - a_i \right\}. \quad (5)$$

If the laws agree, the two derivatives agree on a common interval. Their difference is rational. Multiplying by the product of its denominators produces a polynomial that vanishes on an interval, hence vanishes identically. The rational functions therefore agree everywhere they are defined, including outside the common real domain of the original moment-generating functions.

At the pole  $t = 1/s$ , the residue of (5) is

$$- \sum_{i:a_i=s} \alpha_i.$$

Every summand before the minus sign is positive. Thus the pole cannot cancel, and its residue determines the aggregate shape at  $s$ . Equality of poles and residues gives equality of the measures. This argument concerns the rational logarithmic derivative; it does not treat a Gamma transform with a noninteger shape as meromorphic.

Conversely, the measure determines

$$K_\beta(t) = \int_{\mathbb{R} \setminus \{0\}} [-\log(1 - st) - st] d\mu_\beta(s)$$

on a neighborhood of zero. Equal measures give equal moment-generating functions there and hence equal laws. The empty measure occurs only when every coefficient is zero.  $\square$

Let  $\mu_\beta = \sum_{j=1}^k w_j \delta_{s_j}$ , where the  $s_j$  are distinct and nonzero and  $w_j > 0$ .

**Theorem 2.2** (Exact fiber and minimality). *The fiber  $\mathcal{B}(\beta)$  is in bijection with ordered partitions  $(I_0, I_1, \dots, I_k)$  of  $\{1, \dots, d\}$  satisfying the following conditions:  $I_0$  may be empty, every  $I_j$  for  $1 \leq j \leq k$  is nonempty, and*

$$\sum_{i \in I_j} \alpha_i = w_j, \quad 1 \leq j \leq k. \quad (6)$$

*The corresponding coefficients are*

$$\gamma_i = \begin{cases} 0, & i \in I_0, \\ s_j \sqrt{\alpha_i}, & i \in I_j, \quad 1 \leq j \leq k. \end{cases} \quad (7)$$

*In particular  $|\mathcal{B}(\beta)| \leq (k+1)^d$ . A representation is minimal relative to the fixed dictionary if and only if every  $\beta_i$  is nonzero. In that case  $I_0$  is empty in every alternative and  $|\mathcal{B}(\beta)| \leq k^d$ . When  $k = 0$ , the fiber is  $\{0\}$ .*

*Proof.* Theorem 2.1 forces every nonzero effective scale in an alternative representation to be one of the  $s_j$ , and forces precisely (6). Its converse shows that all such allocations work. Each label has at most  $k+1$  destinations, which proves finiteness.

If a coefficient is zero, that coordinate can be omitted. We allow the empty subvector and empty sum, so this includes the zero response when  $d = 1$ . If every coefficient is nonzero, the total mass of (3) equals  $A = \sum_i \alpha_i$ . Any linear combination of a proper subvector has active measure of total mass at most the sum of that subvector's shapes, which is strictly below  $A$ . It cannot have the same law. The same mass argument shows that every alternative to a full-support vector also has full support.  $\square$

This notion of minimality does not permit replacing the dictionary by different Gamma variables or splitting its components into additional independent variables. For a particular response, labeled uniqueness is equivalent to the existence of exactly one feasible allocation in (6). The uniform criteria below concern all responses for the given dictionary.

### 3 A minimal counterexample

Take  $\alpha = (1, 1, 2)$  and

$$\beta = (1, 1, 2\sqrt{2}), \quad \tilde{\beta} = (2, 2, \sqrt{2}).$$

The effective scales are respectively  $(1, 1, 2)$  and  $(2, 2, 1)$ , so both aggregate measures equal  $2\delta_1 + 2\delta_2$ . More directly, the independent variables

$$A = G_1 + G_2 - 2, \quad B = G_3 - 2$$

are identically distributed. Thus

$$Y_\beta = A + 2B \stackrel{d}{=} 2A + B = Y_{\tilde{\beta}}, \quad M_{Y_\beta}(t) = \frac{e^{-6t}}{(1-t)^2(1-2t)^2}, \quad t < \frac{1}{2}.$$

The squared coefficient multisets are  $\{\{1, 1, 8\}\}$  and  $\{\{4, 4, 2\}\}$ , so no combination of signs and permutations relates the vectors. Both have full support and hence are minimal by Theorem 2.2. The only subsets of the dictionary with total shape 2 are  $\{1, 2\}$  and  $\{3\}$ . Because all the available shape is used, there can be no zero block. Consequently

$$\mathcal{B}(\beta) = \{(1, 1, 2\sqrt{2}), (2, 2, \sqrt{2})\}.$$

This contradicts the literal signed-permutation conclusion in the Discussion of [3], version 1. It does not contradict finite identifiability, called weak identifiability there. The Gamma convolution relation underlying the example is already recognized in that paper; the full-support allocation makes the minimality issue explicit.

### 4 A sharp arithmetic replacement

Let  $H_\alpha$  be the group of permutations preserving the shapes:  $\pi \in H_\alpha$  if  $\alpha_{\pi(i)} = \alpha_i$  for every  $i$ . Its action on coefficients is  $(\pi\beta)_i = \beta_{\pi(i)}$ . These permutations are actual symmetries of the joint predictor law; arbitrary sign flips need not be.

Consider the condition

$$\sum_{i \in I} \alpha_i = \sum_{i \in J} \alpha_i \implies \{\{\alpha_i : i \in I\}\} = \{\{\alpha_j : j \in J\}\} \quad (I, J \subseteq \{1, \dots, d\}). \quad (\text{SR})$$

Equality here is equality of multisets, with multiplicities. The subsets need not contain the same labels.

**Theorem 4.1** (Sharp uniform criterion). *For a fixed positive shape dictionary, the following statements are equivalent:*

- (i) Condition (SR) holds.
- (ii) For every  $\beta \in \mathbb{R}^d$ ,  $\mathcal{B}(\beta) = H_\alpha\beta$ .
- (iii) For every minimal  $\beta$ , every  $\gamma \in \mathcal{B}(\beta)$  is a signed permutation of  $\beta$ .

*Proof.* Suppose (SR) holds. Equivalent representations assign equal total shape to each nonzero scale. By (SR), the constituent shape multisets at that scale agree. Match equal-shape indices within each scale block. The unused indices also have matching shape multisets because the full dictionary is fixed. These matches give a permutation in  $H_\alpha$ . Conversely, permuting equal-shape independent coordinates leaves their joint law unchanged. This proves (i) $\Rightarrow$ (ii), and (ii) $\Rightarrow$ (iii) is immediate.

For the contrapositive of (iii) $\Rightarrow$ (i), suppose (SR) fails. Remove the common indices from a witnessing pair of subsets. The resulting sets  $U, V$  are disjoint, nonempty, have equal total shape, and have different shape multisets. Write  $C$  for their complement and set

$$M = \frac{\max_i \alpha_i}{\min_i \alpha_i}.$$

Choose  $q > \sqrt{M}$  and, if  $C \neq \emptyset$ , choose  $h > q\sqrt{M}$ . Define two effective-scale vectors by

$$(a_i, b_i) = \begin{cases} (1, q), & i \in U, \\ (q, 1), & i \in V, \\ (h, h), & i \in C. \end{cases}$$

Their aggregate measures agree because  $U, V$  have equal total shape. Every scale is nonzero, so both representations are minimal. The squared coefficients at scales  $1, q, h$  lie in pairwise disjoint intervals. In the lowest interval their multisets are exactly the shape multisets of  $U$  and  $V$ , which differ. Thus the complete squared coefficient multisets differ, excluding signed-permutation equivalence.  $\square$

If  $\lambda_1, \dots, \lambda_r$  are the distinct shapes and  $m_1, \dots, m_r$  their multiplicities, condition (SR) is equivalently injectivity of

$$(n_1, \dots, n_r) \mapsto \sum_{\ell=1}^r n_\ell \lambda_\ell \quad \text{on} \quad \prod_{\ell=1}^r \{0, \dots, m_\ell\}. \quad (8)$$

It is also equivalent to the absence of a nonzero integer vector  $z$  with  $|z_\ell| \leq m_\ell$  and  $\sum_\ell z_\ell \lambda_\ell = 0$ . To see the converse formulation, use the positive and negative parts of  $z$  as subset multiplicities.

Repeated shapes are compatible with (SR): every iid Gamma dictionary satisfies it, as does  $(1, 1, 3)$ . Rational independence of the distinct shape values is sufficient but unnecessary. Pairwise distinct shapes alone are insufficient: for  $(1, 2, 3)$  the allocations  $(1, 1, 2)$  and  $(2, 2, 1)$  have equal measures, while their squared coefficient multisets are  $\{\{1, 2, 12\}\}$  and  $\{\{4, 8, 3\}\}$ .

**Corollary 4.2** (Labeled uniqueness). *The map  $\beta \mapsto \mathcal{L}(Y_\beta)$  is injective on  $\mathbb{R}^d$  if and only if all labeled subset sums of  $(\alpha_i)$  are distinct. The same condition is necessary and sufficient if injectivity is demanded only among minimal representations.*

*Proof.* If labeled subset sums are distinct, the mass at every scale identifies its index block uniquely. Conversely, distinct equal-sum subsets reduce, after deleting their intersection, to disjoint nonempty equal-sum subsets. Swap two distinct nonzero scales on these subsets and keep nonzero scales on the complement. The two labeled vectors differ, have full support, and have the same aggregate measure.  $\square$

The distinct-labeled-subset-sum hypothesis is the same arithmetic hypothesis used in the mixed Gamma–Gaussian theorem of [3, Theorem 5]. Corollary 4.2 treats a centered, standardized Gamma dictionary and proves necessity as well as sufficiency there. Theorem 4.1 permits precisely those subset collisions that preserve shape multiplicities. Neither statement is an unqualified strengthening of the entire mixed-model theorem, whose Gaussian mean and original Gamma normalization differ.

**Corollary 4.3** (Generic identification). *For arbitrary positive shapes, if all  $a_i$  are nonzero and pairwise distinct, then  $\mathcal{B}(\beta) = H_\alpha \beta$ . This conclusion therefore holds for Lebesgue-almost every  $\beta \in \mathbb{R}^d$ .*

*Proof.* There are  $d$  distinct observed nonzero scales and only  $d$  predictors. Every scale must receive a predictor, so all blocks are singletons and there is no zero block. Each mass identifies the singleton's shape. The excluded vectors lie in the finite union of hyperplanes  $\beta_i = 0$  and  $\beta_i/\sqrt{\alpha_i} = \beta_j/\sqrt{\alpha_j}$ .  $\square$

## 5 Finite cumulants and an optimal cutoff

Write  $\kappa_n(Y)$  for the  $n$ th cumulant of  $Y$ . For the model (1), differentiation of (4) gives

$$c_n(\beta) := \frac{\kappa_n(Y_\beta)}{(n-1)!} = \sum_i \alpha_i a_i^n, \quad n \geq 2. \quad (9)$$

The first cumulant is always zero.

**Theorem 5.1** (Finite cumulant characterization). *For any two vectors in a fixed  $d$ -predictor Gamma dictionary, equality of their response cumulants of orders  $2, \dots, 2d+1$  is equivalent to equality of their response laws.*

*Proof.* Define

$$\nu_\beta = \sum_{i:a_i \neq 0} \alpha_i a_i^2 \delta_{a_i}.$$

This positive measure has at most  $d$  atoms, and its moments of orders  $m = 0, \dots, 2d-1$  are  $c_{m+2}(\beta)$ . If these moments agree for two vectors, the difference of the measures is supported at  $r \leq 2d$  distinct points  $x_1, \dots, x_r$ . Write its weights as  $u_j$ . Its first  $r$  zero moments give

$$\sum_{j=1}^r u_j x_j^m = 0, \quad 0 \leq m \leq r-1.$$

The square Vandermonde matrix is invertible, so all  $u_j$  vanish. Division of the weight at every nonzero atom by the square of that atom recovers  $\mu_\beta$ . Apply Theorem 2.1. Equality of laws implies equality of cumulants, proving the reverse direction.  $\square$

### 5.1 Exact reconstruction

Let  $m_\ell = c_{\ell+2}(\beta)$  and form the Hankel matrix

$$H_d = (m_{p+q})_{p,q=0}^{d-1}.$$

If  $\mu_\beta = \sum_{j=1}^k w_j \delta_{s_j}$ , then

$$H_d = V \operatorname{diag}(w_1 s_1^2, \dots, w_k s_k^2) V^\top, \quad V_{p,j} = s_j^p.$$

Positive weights and the Vandermonde rank imply  $\operatorname{rank} H_d = k$ . If  $k = 0$ , the response is zero. Otherwise the leading  $k \times k$  block is positive definite. Solve

$$\sum_{j=0}^{k-1} q_j m_{\ell+j} = -m_{\ell+k}, \quad 0 \leq \ell < k. \quad (10)$$

The monic polynomial  $Q(z) = z^k + \sum_{j=0}^{k-1} q_j z^j$  vanishes at each  $s_j$ : equation (10) is a Vandermonde system for the quantities  $w_j s_j^2 Q(s_j)$ . Recover the weights  $w_j s_j^2$  from the first  $k$  moments, divide by  $s_j^2$ , and solve the finite allocation equations (6).

For the counterexample,  $(c_2, c_3, c_4, c_5) = (10, 18, 34, 66)$  and

$$H_2 = \begin{pmatrix} 10 & 18 \\ 18 & 34 \end{pmatrix}, \quad \det H_2 = 16, \quad Q(z) = z^2 - 3z + 2.$$

The roots are 1, 2; the weights of  $\nu$  are 2, 8; and the weights of  $\mu$  are 2, 2. The allocation step yields the two coefficient vectors already displayed.

This is exact population reconstruction, not a stability or efficient-computation guarantee. The finite-atomic stage is a standard Prony construction [5], and near-colliding nodes can amplify perturbations [6]. Exact allocation for arbitrary real shapes requires an exact number representation or equality oracle. Estimated cumulants and approximate subset sums require a separate statistical analysis.

## 5.2 Sharpness of the highest required order

**Theorem 5.2** (Optimal consecutive cumulant cutoff). *For every  $d \geq 1$ , there is a fixed positive Gamma shape dictionary and two full-support coefficient vectors whose response cumulants agree through order  $2d$  and differ at order  $2d + 1$ . Consequently  $2d + 1$  is the smallest uniform highest order in a consecutive initial cumulant segment that determines the response law over all such dictionaries with signed coefficients. The lower-bound examples can have integer shapes.*

*Proof.* For  $1 \leq i \leq d$ , set

$$\alpha_i = \frac{2}{i(d-i)!(d+i)!}, \quad a_i = (-1)^{d-i} i, \quad \beta_i = \sqrt{\alpha_i} a_i, \quad \tilde{\beta}_i = -\beta_i. \quad (11)$$

All shapes are positive and all coefficients are nonzero. For

$$Q(z) = \prod_{j=1}^d (z - j^2)$$

we have

$$Q'(i^2) = \prod_{j \neq i} (i - j)(i + j) = \frac{(-1)^{d-i} (d-i)!(d+i)!}{2i^2},$$

$$\alpha_i a_i^3 = \frac{1}{Q'(i^2)}.$$

For  $0 \leq k \leq d-1$ , Lagrange interpolation of  $z^k$  at the  $d$  nodes  $i^2$ , followed by comparison of the coefficient of  $z^{d-1}$ , gives

$$\sum_{i=1}^d \frac{(i^2)^k}{Q'(i^2)} = \begin{cases} 0, & 0 \leq k \leq d-2, \\ 1, & k = d-1. \end{cases} \quad (12)$$

The first case is empty when  $d = 1$ . Equations (11)–(12) imply

$$\sum_i \alpha_i a_i^{2k+3} = 0 \quad (0 \leq k \leq d-2), \quad \sum_i \alpha_i a_i^{2d+1} = 1.$$

Reflection preserves every even cumulant. The odd cumulants of orders  $3, 5, \dots, 2d-1$  vanish by the preceding identity, and both means vanish by centering. At the next order,

$$\kappa_{2d+1}(Y_\beta) = (2d)!, \quad \kappa_{2d+1}(Y_{\tilde{\beta}}) = -(2d)!.$$

Thus the two laws are different despite agreement through order  $2d$ . Their representations are minimal by Theorem 2.2.

Finally, choose a positive integer  $L$  clearing all denominators of the rational shapes in (11). Replace  $\alpha_i$  by  $L\alpha_i$ , retain the effective scales  $a_i$ , and recompute the coefficients as  $\sqrt{L\alpha_i} a_i$ . Every normalized cumulant is multiplied by  $L$ , preserving the cancellation and the first discrepancy. This gives integer-shape examples in the same dimension.  $\square$

*Remark 5.3* (Meaning of sharpness). The theorem concerns reconstruction of the response law and hence its full coefficient fiber from a consecutive initial cumulant segment. It does not claim optimality for every dictionary, for nonnegative coefficients, for arbitrary nonconsecutive cumulant selections, or for identification only modulo signs. The reflected examples belong to different distributional fibers.

For  $d = 3$ , multiplying the shapes in (11) by 1080 gives

$$\alpha = (45, 9, 1), \quad a = (1, -2, 3), \quad \beta = (3\sqrt{5}, -6, 3), \quad \tilde{\beta} = -\beta.$$

The first distinguishing cumulant has order seven:

| Order | $\kappa_n(Y_\beta)$ | $\kappa_n(Y_{\tilde{\beta}})$ |
|-------|---------------------|-------------------------------|
| 2     | 90                  | 90                            |
| 3     | 0                   | 0                             |
| 4     | 1,620               | 1,620                         |
| 5     | 0                   | 0                             |
| 6     | 162,000             | 162,000                       |
| 7     | 777,600             | -777,600                      |

The variance also equals  $\|\beta\|^2 = 45 + 36 + 9 = 90$ .

## 6 Extensions

### 6.1 One Gaussian predictor

**Proposition 6.1.** *Let  $Z \sim N(0, 1)$  be independent of the Gamma dictionary. For  $c, e \in \mathbb{R}$ ,*

$$Y_\beta + cZ \stackrel{d}{=} Y_\gamma + eZ \iff \mu_\beta = \mu_\gamma \text{ and } c^2 = e^2.$$

*Proof.* The logarithmic derivative of the joint transform is (5) plus  $c^2 t$ . Comparing poles and residues again identifies  $\mu_\beta$ , since the added polynomial has no poles. Subtracting the now identical Gamma terms identifies  $c^2$ . The converse follows from equality of the transforms near zero.  $\square$

Under (SR), the only ambiguities are equal-shape Gamma permutations and the Gaussian sign. Every fiber is finite even without (SR). For several independent standard Gaussian coordinates, the Gaussian part determines only the squared coefficient norm; a nonzero norm then gives a continuous ambiguity. If a single Gaussian has known mean  $\eta \neq 0$  and variance  $\sigma^2 > 0$ , its coefficient is recovered from the remaining mean after the Gamma part is determined.

## 6.2 An analytic convolution-family theorem

**Theorem 6.2.** *Suppose a known family  $(Z_\alpha)_{\alpha>0}$  has moment-generating functions on neighborhoods of zero satisfying*

$$\log \mathbb{E} e^{tZ_\alpha} = \alpha\psi(t),$$

*where  $\psi$  is analytic,  $\psi(0) = 0$ , and  $v = \psi''(0) > 0$ . Fix  $d$  and positive  $\alpha_1, \dots, \alpha_d$ , take independent  $Z_{\alpha_i}$ , and put*

$$X_i = \frac{Z_{\alpha_i} - \alpha_i\psi'(0)}{\sqrt{\alpha_i v}}, \quad a_i = \frac{\beta_i}{\sqrt{\alpha_i v}}. \quad (13)$$

*Assume  $\psi^{(n)}(0) \neq 0$  for  $n = 2, \dots, 2d + 1$ . Then equality of response laws is equivalent to equality of  $\sum_{a_i \neq 0} \alpha_i \delta_{a_i}$ , and is determined by cumulants of those orders. The fiber, minimality, arithmetic-criterion, labeled-uniqueness, and generic-identification conclusions above continue to hold, with coefficient reconstruction*

$$\gamma_i = s_j \sqrt{\alpha_i v} \quad (i \in I_j), \quad \gamma_i = 0 \quad (i \in I_0).$$

*Proof.* The response cumulant-generating function is

$$K_\beta(t) = \sum_i \alpha_i [\psi(a_i t) - a_i t \psi'(0)].$$

For every required order,

$$\frac{\kappa_n(Y_\beta)}{\psi^{(n)}(0)} = \sum_i \alpha_i a_i^n.$$

The nonvanishing assumption permits division. The Vandermonde argument of Theorem 5.1 recovers the aggregate measure from these normalized cumulants. Conversely, that measure determines  $K_\beta$  locally. The finite allocation and total-positive-mass arguments therefore apply unchanged. For the necessity part of Theorem 4.1, the squared coefficients are now  $\alpha_i v a_i^2$ ; the common positive factor  $v$  preserves the separated-interval construction.  $\square$

The assumptions are sufficient and are not asserted to be necessary. A finite set of nonzero derivatives suffices for a fixed  $d$ . Gamma variables have  $\psi(t) = -\log(1-t)$  and  $v = 1$ . Poisson variables have  $\psi(t) = e^t - 1$  and  $v = 1$ , so every required derivative is nonzero. Pure Gaussian convolution families have zero higher cumulants and do not satisfy this hypothesis. No worst-case cutoff claim for each individual convolution family is inferred from the Gamma lower bound.

## 6.3 Direct verification for Poisson predictors

For independent  $P_i \sim \text{Poisson}(\lambda_i)$  and  $X_i = (P_i - \lambda_i)/\sqrt{\lambda_i}$ , put  $a_i = \beta_i/\sqrt{\lambda_i}$ . Then

$$K''_\beta(t) = \sum_{a_i \neq 0} \lambda_i a_i^2 e^{a_i t}.$$

Distinct real exponentials are linearly independent: if their linear combination vanishes near zero, its successive derivatives at zero give an invertible Vandermonde system. Thus  $K''_\beta$  determines the scales and the weights  $\lambda_i a_i^2$ . Division by  $a_i^2$  recovers  $\sum_{a_i \neq 0} \lambda_i \delta_{a_i}$ . Equality of this measure gives equality of the centered transforms. This supplies a direct proof of the Poisson case without Gamma pole calculations.

## 7 Scope, reproducibility, and research status

The results distinguish three questions: recovering a distributional aggregate, allocating that aggregate to a fixed predictor dictionary, and reconstructing it from finitely many cumulants. Minimality excludes unused predictors but does not guarantee unique allocation. Condition (SR) is the exact uniform replacement for signed-permutation identification in the Gamma model. Finite ambiguity holds without that condition.

The unrestricted finite-identifiability question for arbitrary independent standardized non-Gaussian predictors is not resolved here. Neither are finite-sample estimation, noisy-response deconvolution, unknown shape parameters, numerical conditioning guarantees, or efficient exact subset-sum algorithms. The lower bound permits signed coefficients and concerns the highest order of a consecutive cumulant segment.

The accompanying public package contains the manuscript source, original standard-library exact checkers, machine-readable outputs, a response to the supplied review, and provenance records. The original finite grid covers 2,055 scale vectors. Additional polynomial checks test the lower-bound identity in dimensions 1 through 20. Semantic negative controls reject changed signs, shape allocations, cumulant normalization, and a missing variance factor. These finite calculations detect implementation errors; the all-dimensional claims rest on the written proofs.

**Research status and disclosure.** This is an unrefered candidate. AI tools assisted proof development, manuscript preparation, exact-check orchestration, and internal adversarial review. The cumulant lower-bound construction originated in a supplied review and was checked before inclusion. Supplied review identities and processes have not been authenticated; local replay does not establish external independence or endorsement. Historical novelty is not certified, and no proof-assistant formalization is supplied. Anonymous is the scholarly creator under the Evidence Press publication protocol.

**Data and code availability.** All computational inputs are explicit rational constants or formulas in this paper. The public package contains original Python checkers and reproduction instructions. No empirical or personal data are used. Original prose and data are released under CC0-1.0 and original code under MIT. Supplied third-party review files are not redistributed. Repository: <https://github.com/ipitchford/unlinked-gamma-regression>. Version 0.1.0-candidate archive: <https://doi.org/10.5281/zenodo.22858495>.

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